Combining Five Acoustic Level Modeling Methods for Automatic Speaker Age and Gender Recognition

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Introduction

Automatic recognition of paralinguistic information from speech is important.

- Speaker identity, gender, age range, emotional state, etc.
- Guide human computer interaction systems to automatically adapt to different user needs.

Our focus is to enhance the performance of acoustic-level approaches for speaker age and gender identification.

The proposed approach by combining five different acoustic level methods:

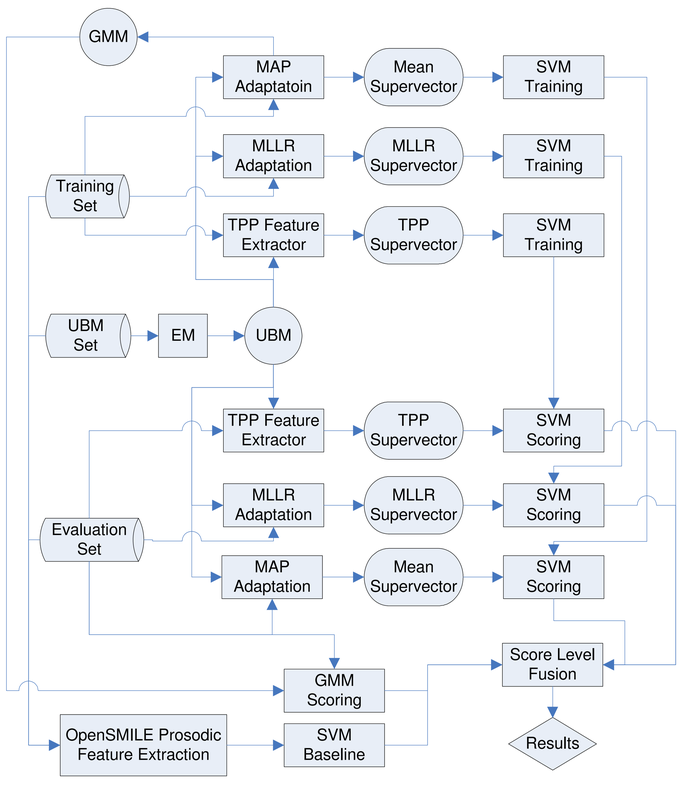
1. GMM baseline
2. GMM-SVM mean supervector (stacked means of MAP adapted models)
3. GMM-SVM maximum likelihood linear regression (MLLR) matrix supervector [1]
4. GMM-SVM Tandem posterior probability (TPP) supervector [2]
5. SVM baseline using acoustics level features (including prosodic features)

Contribution:

1. The GMM-SVM mean supervector method is extended by two kinds of supervectors, MLLR supervector and Tandem posterior probability (TPP) supervector.
2. Combining these acoustic level methods at the score level can improve the overall performance.

Methods

System overview



GMM-SVM mean supervector

- A two stage framework: map the supervectors into discriminative aGender characterization score vectors (DACSV), followed by a back end SVM classifier.

GMM-SVM maximum likelihood linear regression (MLLR) supervector

- MLLR adaptation was performed using UBM model and training set.
- Linear Discriminant Analysis (LDA) was employed for dimension reduction.

GMM-SVM Tandem posterior probability (TPP) supervector

Given a frame of MFCC feature x_t and the GMM-UBM λ:

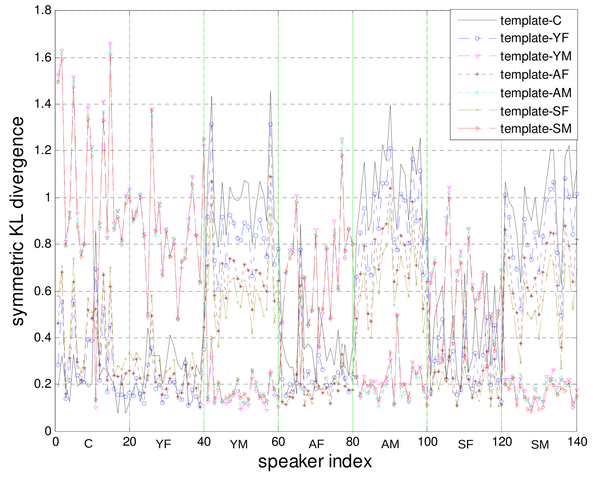
$$\lambda = \{w_i, \mu_i, \Sigma_i\}, i = 1, \cdots, M$$

The posterior occupancy probability is calculated as follows:

$$P(\lambda_i|x_t) = \frac{w_i p_i(x_t|\mu_i, \Sigma_i)}{\Sigma_{j=1}^{M} w_j p_j(x_t|\mu_j, \Sigma_j)}$$

Thus the TPP supervector for a T frames segment is:

$$TPP_{supervector} = [b_1, b_2, \cdots b_M]$$
$$b_i = \frac{1}{T}\Sigma_{t=1}^{T} P(\lambda_i|x_t)$$



Corpus and Classification Task

aGender database [3] is used.

- Speech duration after VAD in the training and development data sets are 1.13 ± 0.86 seconds and 1.14 ± 0.87 seconds, respectively.
- The training data set of the aGender database (471 speakers) was divided into 2 parts: the **training set** (20 speakers for each age/gender group, MAP and MLLR adaptation usage) and the **UBM set** (the rest 331 speakers).
- The development data set from the aGender database (20548 utterances) is adopted as the **evaluation set** in this paper.
- The testing data set from the aGender database (17332 utterances) is evaluated as the official challenge results. (Reported in the end.)

Experimental results

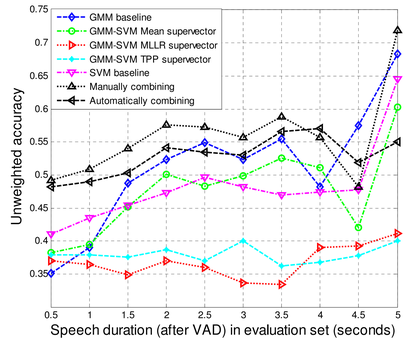
Performance of each method and the fusion approach (MFuse and AFuse denote manually and automatically (inverse entropy) tuned weight in score level fusion.)

	age & gender		age		gender	
ID & System	UA	WA	UA	WA	UA	WA
1.GMM baseline	43.1	42.4	47.0	45.0	76.0	81.7
2.GMM-SVM mean supervector	42.6	43.1	46.0	45.5	75.6	82.8
3.GMM-SVM MLLR supervector	36.2	36.1	40.5	41.1	68.4	75.8
4.GMM-SVM TPP supervector	37.8	38.0	41.6	41.5	71.2	79.6
5.SVM baseline	44.6	45.0	47.4	46.7	77.6	85.2
MFuse 1+2	45.2	45.3	47.4	48.7	77.6	83.7
MFuse 3+4	40.3	40.3	44.1	43.5	73.4	81.0
MFuse 1+2+3+4	50.4	50.5	53.7	51.8	83.7	89.0
MFuse 1+2+3+4+5	**52.7**	**53.0**	**55.5**	**54.0**	**84.7**	**90.3**
AFuse 1+2+3+4+5	**51.2**	**51.4**	**54.6**	**52.5**	**84.7**	**89.7**

Confusion matrix (left) and accuracy for different valid speech durations (right) for 7 class age and gender task.

	C	YF	YM	AF	AM	SF	SM
C	**67**	14	6	5	1	6	1
YF	16	**59**	1	16	0	8	0
YM	0	0	**57**	2	18	3	20
AF	4	25	1	**38**	0	32	0
AM	0	0	35	2	**25**	1	37
SF	5	9	1	28	0	**56**	1
SM	0	0	15	1	16	3	**66**

Valid speech duration in evaluation set are 1.14 ± 0.87 seconds



By using the official testing data set from the challenge, our AFuse system achieved:

- 50.12% UA, 47.52% WA on the age recognition task.
- 82.38% UA, 86.27% WA on the gender recognition task.

Conclusion and future work

Conclusion:

- MLLR supervector and TPP supervector systems perform well in this task.
- Combining different acoustic level methods can enhance the performance.

Future works:

- investigating the GMM-SVM Constrained MLLR supervector method
- combining other prosodic or phonetic level methods

References

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